

Iyed MEDIOUNI

AI ENGINEER · LLM SYSTEMS & APPLIED MACHINE LEARNING

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PROFILE

AI engineer who takes language-model systems all the way into production. I build assistants that answer from a company's own live systems, and the evaluation loop that proves each version is actually better before it ships. The assistant I built during my internship was sold to a client. Comfortable across the whole path: retrieval design, multi-agent orchestration, on-premise inference, FastAPI services and the console the team runs it from.

EXPERIENCE

02/2026 - 07/2026
Lac 1, Tunis

- AI Engineer Intern** - TELNET Holding ● github.com/iyed122/ai-assistant
- Project knowledge sat fragmented across Jira, Confluence and GitLab: delivered a production, on-premise assistant on LangGraph and Neo4j Graph-RAG that answers across all three from one natural-language interface. TELNET sold the product to a client.
 - Responses were too slow to feel interactive: built a hybrid query-routing layer pairing deterministic classification with parallel API dispatch, cutting mean latency 2.4x (7.3s to 3.0s).
 - Fine-tuning ran blind with no measure of answer quality: designed Hammer, an evaluation framework (RAGAS + deterministic validators) grading every answer on a four-tier scale crossed with a failure taxonomy, so no QLoRA/DPO run ships without a proven gain.
 - Traced a silent scoring defect that mislabelled correct answers as failures and was poisoning the training set; the fix lifted training-data quality ~10 points and model faithfulness 27%.
 - Took it from prototype to operable: Docker-containerised FastAPI services, MongoDB persistence, local inference via Ollama, MLflow tracking every fine-tuning run, and a React/Vite console for training mode, DPO curation and QLoRA review - with the evaluation loop doubling as production monitoring for answer quality, latency and failure type.

PROJECTS

07/2025 - 10/2025

- Autonomous Vehicle Perception & Trajectory Prediction** ● github.com/iyed122/autonomous-vehicule-VLM
- Camera detection alone cannot anticipate where an object is heading: built an end-to-end pipeline on AWS SageMaker pairing a fine-tuned YOLOv8 detector with a custom bird's-eye-view model for LiDAR trajectory prediction, trained in PyTorch to ADE 0.8 / FDE 1.2.
 - Made the output legible to non-specialists through a Streamlit dashboard surfacing live detection, trajectories and a CLEAR / CAUTION / BRAKE collision-risk signal across 6-camera and LiDAR views.
 - Integrated a local vision-language model (Qwen2.5-VL) that narrates driving scenes across the full sensor suite, turning raw detections into structured hazard analysis.

09/2025 - 11/2025

- Multi-Agent Content Automation** ● github.com/iyed122/automated-twitter-content
- Generic model output drifts from what is actually trending: built a CrewAI + LLaMA-3.1-8B pipeline where specialised agents research live web, search and trend data before any drafting begins.
 - Closed the loop to publication through the X API with secure OAuth, plus a grounding step tying every generated post back to its retrieved sources.

02/2023 - 06/2023
Ghazela, Technopole
Final-year project

- AURAX** - full-stack commerce platform
- Built the complete customer journey in React and Node.js (MERN): search and filters, cart, secure online payment via Stripe, promotion codes, order confirmation, tracking and history.
 - Added a content-based recommender (VGG16 to PCA to KNN) served through a Flask API, working from a user's first visit with no prior history.

EDUCATION

2023 - 2026

Engineering Degree - Data Science & Artificial Intelligence
TEK-UP University

2020 - 2023

Bachelor in Computer Science
ISAMM, Manouba

TECHNICAL SKILLS

- AI & ML** LangGraph · LangChain · CrewAI · Graph-RAG · RAGAS · QLoRA / DPO fine-tuning · Ollama · PyTorch · TensorFlow · Scikit-learn · OpenCV
- LANGUAGES** Python · JavaScript · SQL · PL/SQL · R · MATLAB
- DATA & BACKEND** Neo4j · Qdrant · MongoDB · PostgreSQL · MySQL · Cassandra · FastAPI · Pandas · NumPy
- CLOUD & MLOPS** AWS SageMaker · EC2 · Lambda · S3 · DynamoDB · Redshift · RDS · Docker · MLflow
- FRONTEND & TOOLING** React · Vite · Node.js · Streamlit · Git / GitLab · LaTeX

LANGUAGES & CERTIFICATIONS

- ARABIC** Native
- FRENCH** Professional working proficiency - academic and technical writing
- ENGLISH** IELTS Academic, British Council - B2 (6.5 / 9)